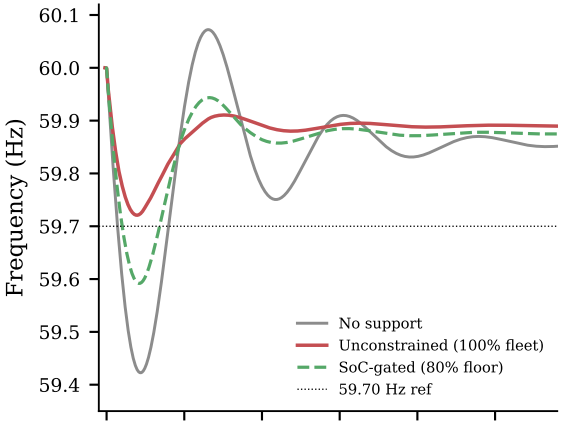
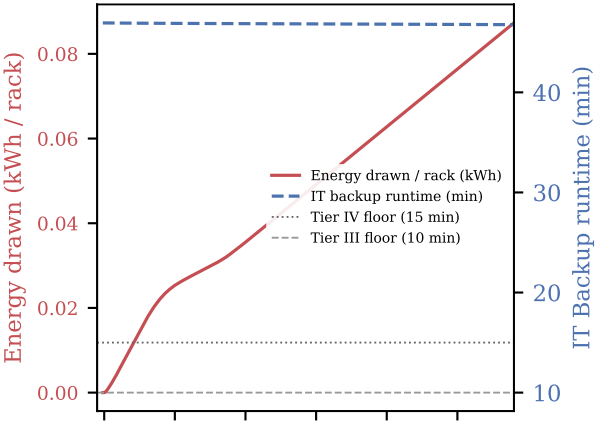

(a) SoC reserve-gated recovery
Frequency (Hz)
60.1
60.0
59.9
59.8
59.7
59.6
59.5
59.4
0
5
10
15
20
25
Time since contingency (s)
No support
Unconstrained (100% fleet)
SoC-gated (80% floor)
59.70 Hz ref


(b) Energy throughput & backup autonomy
Energy drawn (kWh / rack)
0.08
0.06
0.04
0.02
0.00
IT Backup runtime (min)
40
30
20
10
0
5
10
15
20
25
Time since contingency (s)
Energy drawn / rack (kWh)
IT backup runtime (min)
Tier IV floor (15 min)
Tier III floor (10 min)